

**UNITED STATES INTERNATIONAL TRADE COMMISSION
WASHINGTON DC**

**Before the Honorable Clark S. Cheney
Chief Administrative Law Judge**

In the Matter of

**CERTAIN SEMICONDUCTOR DEVICES,
PRODUCTS CONTAINING SAME, AND
COMPONENTS THEREOF**

Investigation No. 337-TA-1500

COMPLAINANT'S RESPONSIVE CLAIM CONSTRUCTION BRIEF

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Respondents’ overarching approach to claim construction is that they cannot understand anything. They cannot understand the patents. They cannot understand GlobalFoundries’ infringement or validity contentions. And so, Respondents incorrectly say, four of the six disputed terms in four different patents are indefinite. But indefiniteness turns on whether the claims, read in light of the specification and prosecution history, inform a skilled artisan of the scope of the invention with reasonable certainty, not on whether a litigant is uncertain about its adversary’s contentions. *See Ironburg Inventions Ltd. v. Valve Corp.*, 64 F.4th 1274, 1290 (Fed. Cir. 2023) (“[D]isputes over infringement do not make a patent claim indefinite.”). Respondents’ remaining two constructions fail because they import preferred embodiments into the claims.

A. ’546 Patent/’167 Patent

1. “layer”/“first liner layer”/“refractory metal layer” (’546 patent at claims 1-14; ’167 patent at claims 1, 3, 4)

Complainant’s Proposed Construction	Respondents’ Proposed Construction
Plain and ordinary meaning	Indefinite
Not indefinite	If not indefinite, “a separate and defined [first liner / refractory metal] coating of material”

Respondents concede that “layer” has a plain and ordinary meaning. Resp. Br. at 4 (“The terms ‘layer,’ ‘first liner layer,’ and ‘refractory metal layer’ have plain-and-ordinary meanings.”). That alone forecloses their indefiniteness challenge. Respondents nonetheless contend that “layer” requires construction and should be construed as “a separate and defined [first liner / refractory metal] coating of material.” That construction is unsupported. The only technical dictionary Respondents cite defines “layer” as “[a] defined thickness which is part of a material, or which surrounds it.” Resp. Br. at 4. That definition contains no requirement that a layer be “separate” or limited to a “coating of material.” Respondents’ proposed construction should be rejected.

Respondents confirm that their proposed construction is intended to limit “layer” to a *single* coating of material, thereby excluding sublayers. The intrinsic evidence precludes that limitation. For example, claim 4 of the ’546 patent recites a “refractory metal layer” that includes “a bilayer of titanium (Ti) and titanium nitride (TiN).” *See also* ’546 patent at cls. 7, 11, 14. The specification similarly teaches a “dielectric layer” that “may include a single layer *or more than two separate layers*” (’546 patent, 4:56-62) and that the “refractory metal layer” may be a “Ti/TiN bilayer.” *Id.*, 7:45-46. Respondents’ proposed construction would read out not only disclosed embodiments, but dependent claims as well. It must be rejected.

Respondents resort to arguing that unless “layer” is limited to a single layer, the term is indefinite. Respondents are wrong. As an initial matter, Respondents do not cite a single case where a court has found the term “layer” indefinite. And in every case Respondents cite, the court construed “layer,” confirming it is not indefinite. Respondents contend that GlobalFoundries’ infringement and domestic industry contentions support indefiniteness. But that argument invites legal error. “The test for indefiniteness does not depend on a potential infringer’s ability to ascertain the nature of its own accused product to determine infringement, but instead on whether the claim delineates to a skilled artisan the bounds of the invention.” *Ironburg*, 64 F.4th at 1290 (quoting *SmithKline Beecham Corp. v. Apotex Corp.*, 403 F.3d 1331, 1340-41 (Fed. Cir. 2005)). Such disputes “are disputes about whether there is infringement, not disputes about whether the patent claims are indefinite.” *Id.* (quoting *Presidio Components, Inc. v. Am. Tech. Ceramics Corp.*, 875 F.3d 1369, 1377 (Fed. Cir. 2017)). Respondents’ reliance on GlobalFoundries’ contentions to argue indefiniteness should be disregarded accordingly.

Substantively, Respondents mischaracterize GlobalFoundries’ contentions, and every criticism reduces to two propositions: (1) that a “layer” cannot include sublayers, and (2) that a

single accused structure cannot supply material for more than one recited element. Resp. Br. at 5-6; Banerjee Decl. ¶¶ 56, 64-66. Neither is correct. As shown above, the claims and specification expressly contemplate that a “layer” may comprise sublayers. And nothing in the claims forbids one deposited layer from satisfying, in part, more than one limitation. For example, Respondents argue Complainant “isolate[s] a portion of the tungsten metal filler” as the “refractory metal layer” and “the rest of the tungsten metal filler” as the “metal substantially filling the contact opening” limitation.” Resp. Br. at 7. But whether a particular titanium or tungsten film can comprise sublayers or corresponds to the “refractory metal layer” rather than the “metal substantially filling the contact opening” is a quintessential factual dispute about infringement; it is not a basis to find the claims indefinite. *Ironburg*, 64 F.4th at 1290.

2. “intermix region” (’546 patent at claims 1, 2, 6, 8-9, 13; ’167 patent at claims 1, 2, 6)

Complainant’s Proposed Construction	Respondents’ Proposed Construction
Plain and ordinary meaning	Indefinite
Not indefinite	

Respondents contend that “intermix region” is indefinite without offering any alternative construction, effectively arguing the term is beyond comprehension. Respondents are wrong.

The claims and specification inform a POSITA what an intermix region is, where it is located, and what it contains. Claim 1 of the ’546 patent recites “causing a portion of the first liner layer to diffuse into the upper portion of the semiconductor substrate to form a first intermix region at the upper portion of the semiconductor substrate.” Claim 1 of the ’167 patent recites “an intermix region at an upper surface of the semiconductor substrate in the contact opening, the intermix region having a composition including a material used in the first liner layer, a material used in the refractory metal layer, and a material used in the semiconductor substrate.” The

specification explains that intermix regions “originate from the high energy of the incoming sputtered atoms (from first liner layer 132) which are thus pushed into the Si lattice,” creating “a thin region . . . of mixed Si and metal on the order of 3-5 nm in thickness” and gives exemplary compositions, including “a NiPt silicide,” “a NiPtTi silicide,” “a Ni silicide,” or “a NiTi silicide.” ’546 patent, 6:3-19, 7:45-58. The dependent claims add that “the intermix region includes a titanium-based silicide” and “has a thickness of approximately 3 nanometers to approximately 5 nanometers.” ’167 patent at cls. 2, 6. *See* GF Opening Br. at 9-10; Liu Decl. ¶¶ 39-42.

Ignoring the intrinsic evidence, Respondents rely on a figure from a 2004 article on laser thermal processing of titanium/cobalt bilayers, **nine years** before the ’546 patent was filed, to argue a POSITA cannot tell whether an intermix region is present. Resp. Br. at 11-12 (citing Ex. 15). This argument invites legal error. First, the article has nothing to do with the asserted patents or any accused product; it is not even from the same time frame as the ’546 patent. Second, “when the disputed terms can be understood from a careful reading of the public record,” “reliance on [prior art] is unnecessary and indeed improper” and cannot be used to “vary claim terms from how they are defined, even implicitly, in the specification or file history.” *Vitronics Corp. v. Conceptronic, Inc.*, 90 F.3d 1576, 1584 (Fed. Cir. 1996).

Respondents’ remaining arguments are that a POSITA would not know the amount of intermixing required, whether the intermix region must be intentional, or whether it must be “uniform.” Each of these arguments fails. First, the patents need not specify an “amount” of intermixing. “[A]bsolute precision” in claim language is “unattainable.” *Interval Licensing LLC v. AOL, Inc.*, 766 F.3d 1364, 1370 (Fed. Cir. 2014). Nor does the law demand “mathematical precision to comply with the definiteness requirement.” *Sonix Tech. Co. v. Publ’ns Int’l Ltd.*, 844 F.3d 1370, 1377 (Fed. Cir. 2017); *Ironburg*, 64 F.4th at 1285 (“elongate member” definite despite

absence in specification of “objective guidance as to how much longer than wide the member must be”).

Second, the patents need not specify whether the intermixing is purposeful or incidental, as it is error to “read a ‘purpose’ requirement” into claims reciting “structural limitations,” *Toshiba Corp. v. Imation Corp.*, 681 F.3d 1358, 1368 (Fed. Cir. 2012), or to “limit claim language to exclude particular devices because they do not serve a perceived ‘purpose’ of the invention,” *E-Pass Techs., Inc. v. 3Com Corp.*, 343 F.3d 1364, 1370 (Fed. Cir. 2003); accord *Praxair, Inc. v. ATMI, Inc.*, 543 F.3d 1306, 1325 (Fed. Cir. 2008). Respondents’ contention thus improperly asks the CALJ to import a limitation into the claims.

Third, the claim language answers Respondents’ contentions regarding uniformity, imposing no such requirement. Indeed, courts have found uniformity is not needed when absent from the claims. *See, e.g., McNeil-PPC, Inc. v. Perrigo Co.*, 443 F. Supp. 2d 492, 503-07 (S.D.N.Y. 2006), *on reconsideration in part*, 2007 WL 104513 (S.D.N.Y. 2007) (rejecting uniformity limitation on a mixture unsupported by intrinsic evidence); *Gemak Trust v. Reckitt Benckiser LLC*, 2020 WL 4284973, at *4 (D. Del. 2020) (same). The specification confirms the lack of any required uniformity, disclosing second intermix regions containing “a mixture of the materials used for source and drain regions 124a, 124b, first liner layer 132, and refractory metal layer 140.” ’546 patent, 8:10-20.

The term “intermix region” is not indefinite and should be given its plain and ordinary meaning.

B. ’235 Patent

1. “from a nickel silicide surface at a silicon interface to an opposite surface of the nickel silicide” (’235 patent at claims 1, 10)

Complainant’s Proposed Construction	Respondents’ Proposed Construction
Plain and ordinary meaning	Indefinite

Not indefinite	
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Respondents do not dispute that a POSITA would be familiar with the terms “nickel silicide surface,” “silicon interface,” and “opposite surface.” Resp. Br. at 15. Instead, Respondents’ indefiniteness argument is that a POSITA would not understand *where* the claimed “silicon interface” lies. Respondents are wrong.

Claim 1 first recites a “silicon substrate” and a “nickel silicide layer.” Claim 1 then specifies that the “nickel silicide layer” includes a “gradient” from “a nickel silicide surface *at a silicon interface* to an opposite surface of the nickel silicide.” ’235 patent cl. 1. A POSITA would readily understand “a nickel silicide surface at a silicon interface” as a boundary between the nickel silicide and the silicon below. Ex. 9 (1995 Webster’s II Dictionary) (defining interface as “a surface forming a common boundary between adjacent regions”); see GF Opening Br. at 10-11; Liu Decl. ¶¶ 43-48.

The Federal Circuit’s decision in *Ironburg* governs. There, the claims recited an elongate member “that extends substantially the full distance between the top edge and the bottom edge.” *Ironburg*, 64 F.4th at 1282. The defendant did not dispute that a controller comprises a “top edge” and a “bottom edge,” instead arguing skilled artisans “would not know where to locate the ‘bottom edge’” and “would not know *where* to measure” the distance between the top and bottom edges. *Id.* at 1290. The Court rejected that argument, because such a dispute merely “g[ave] rise to a question of infringement,” not indefiniteness. *Id.* The same result follows here. The claims recite two physical endpoints of the same nickel silicide layer: the silicon-facing surface and the opposite surface. The parties here agree on what to measure—the platinum percentage within the nickel silicide layer—and disagree only as to *where* within the layer to take that measurement. Such a dispute is insufficient to sustain an indefiniteness challenge. *Ironburg*, 64 F.4th at 1290

(“Reasonable disputes over infringement do not make a patent claim indefinite.”); *see also Certain Oil Vaporizing Devices, Components Thereof, and Products Containing the Same*, Inv. 337-TA-1392, Order No. 19, Construing Claim Terms, at 31 (Aug. 30, 2024) (rejecting indefiniteness challenge premised on “multiple methodologies for determining the physical boundary” of a claimed “bottom portion,” because “[t]he fact that Respondents disagree about the scope of the term is not enough to render it indefinite”).

Respondents’ reliance on *Teva* and *Dow* is misplaced. Each concerned a dispute over *how* to measure a numerical value—“molecular weight” in *Teva* and “slope of strain hardening coefficient” in *Dow*—where multiple competing methodologies existed for computing the claimed value and each methodology produced a different number. *Teva Pharms. USA, Inc. v. Sandoz, Inc.*, 789 F.3d 1335, 1341 (Fed. Cir. 2015) (“The parties agree that ‘molecular weight’ could refer to Mp, Mw, or Mn. And they agree that each of these measures is calculated in a different way and would typically yield a different result.”); *Dow Chemical Co. v. Nova Chemicals Corp.*, 803 F.3d 620, 633-35 (Fed. Cir. 2015) (at least three established methods for measuring slope of strain hardening, and the patentee’s expert devised a fourth). The defect in those patents was the failure to teach *how* to perform the calculation, not *where* to perform it. *Dow*, 803 F.3d at 634 (“[T]he existence of multiple methods leading to different results without guidance in the patent or the prosecution history as to which method should be used renders the claims indefinite.”). Moreover, in *Dow*, the claim term at issue—“slope of strain hardening coefficient”—was “a new Dow construct, not previously known in the art.” *Id.* at 631. Here, by contrast, Respondents concede that “nickel silicide surface” and “silicon interface” are “not necessarily unfamiliar” to a POSITA. Resp. Br. at 15.

The Court in *Ironburg* rejected the same comparison Respondents press here. The Court held that the challenger’s “weakness” in *Ironburg* and “a principal distinction from *Dow* and *Teva*,” was that “the parties agree on how to measure . . . and disagree as to where to find those two points.” *Ironburg*, 64 F.4th at 1291. That, the Court explained, “is essentially the opposite of *Dow*, where we assumed the patentee was correct as to where to take the pertinent measurement . . . but invalidated the claims due to the uncertainty as to *how* to measure.” *Id.* So too here. Respondents do not dispute a “silicon interface” exists between nickel silicide and silicon; they dispute where it is. That is a “dispute[] about whether there is infringement, not [a] dispute[] about whether the patent claims are indefinite.” *Id.* at 1290 (quoting *Presidio*, 875 F.3d at 1377); *see also SmithKline Beecham*, 403 F.3d at 1340-41.

Respondents attempt to manufacture a dispute about *how* to measure so as to fit these facts to *Teva* and *Dow*. It fails. Respondents identify two techniques for locating an “interface” in a semiconductor device: the Adusumilli method and the analysis of electron refraction patterns. Resp. Br. at 15-16; Ernst Decl. ¶¶ 53-54, 56-57. But those are techniques for identifying *where* to measure the platinum concentration within the nickel silicide layer, not *how* to measure it.

Respondents then suggest that different measurement techniques “would yield different results.” That is unsupported. It rests entirely on two lines, “A” and “B,” that Respondents themselves arbitrarily drew. Resp. Br. at 18. But Respondents nowhere contend that they applied the Adusumilli method, the electron refraction method, or any technique at all in placing those lines. Nor do they stand behind the underlying data, stating that they “do not adopt any of the values listed” and “disagree that these figures are representative of the amount of any element in Respondents’ products.” Resp. Br. at 18 n.4. Drawing two arbitrary lines and observing that there would be no infringement if the silicon interface happened to lie at those points establishes nothing

about whether competing methodologies yield different results. Any suggestion that Lines A and B are the result of different techniques is misleading at best.

Disputes over “the proper application of the test methodology in the circumstances of an individual case” go to infringement, not indefiniteness. *Presidio*, 875 F.3d at 1377; *see also Barrette Outdoor Living, Inc. v. Fortress Iron, LP*, 156 F.4th 1353, 1367 (Fed. Cir. 2025). Accepting Respondents’ approach “would allow an infringer to manufacture indefiniteness by simply hiring an expert willing to disagree with the patentee’s infringement analysis.” *Ironburg*, 64 F.4th at 1290. Respondents’ alternative request that the CALJ “defer consideration of indefiniteness until after the parties have developed the record during fact and expert discovery” confirms as much. Resp. Br. at 39. Respondents’ complaint is not that the claims fail to inform a POSITA of the scope of the invention with reasonable certainty; it is that they would prefer to litigate the location of the silicon interface on a fuller infringement record.

C. ‘983 Patent

1. **“adjacent to” (‘983 patent at claims 1, 5, 13, 17, 20)/“the channel is displaced from [a/the] first heavily doped region adjacent to the first edge of the gate” (‘983 patent at claims 1, 13, 20)**

Complainant’s Proposed Construction	Respondents’ Proposed Construction
Plain and ordinary meaning	Indefinite
Not indefinite	

Respondents’ brief confirms that this term is not indefinite and should be accorded its plain meaning, which is “situated near or next.” Ex. 8 (2004 Merriam-Webster Dictionary) at 4.

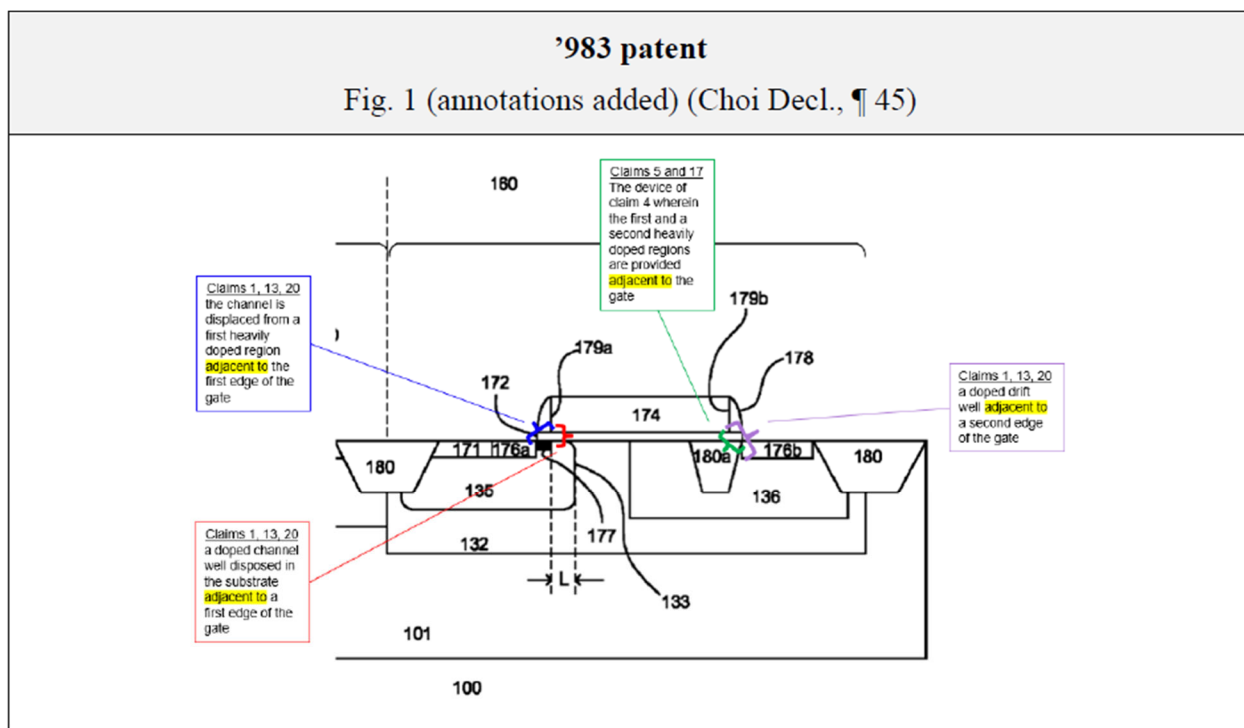
The term “adjacent” has consistently been found not indefinite by courts. *E.g., Smith Interface Techs., LLC v. Apple Inc.*, 2025 WL 3305622, at *6-*7 (S.D. Cal. Oct. 3, 2025) (rejecting defendant’s indefiniteness argument for “adjacent” as improperly requiring “mathematical precision”); *Denneroll Holdings Pty Ltd. v. Chirodesign Grp., LLC*, 2016 WL 705207, at *5 (S.D.

Tex. Feb. 23, 2016) (similar). Respondents do not provide a single counterexample of “adjacent” ever being found indefinite. Instead, Respondents assert that “[t]he claims recite elements that are ‘adjacent to’ each other across different distances and despite the presence of other elements in between.” Resp. Br. at 20. Respondents’ complaint is meritless.

The meaning of relational terms is “dictated by context,” *Accentra, Inc. v. Staples, Inc.*, 500 F. App’x 922, 930-31 (Fed. Cir. 2013), and the intrinsic record supplies sufficient context. Claim 1 uses clear, well-understood terms to provide a POSITA with reasonable certainty about the locations of the various structures. In claim 1, adjacency is measured against a specific reference feature: the “first edge” or the “second edge” of the gate. The “doped channel well” and “heavily doped region” are both “adjacent to” the first edge, and the “doped drift well” is “adjacent to” the second edge. Claim 5 further specifies that the heavily doped region is disposed “in the doped channel well on a source side of the device,” which is perfectly consistent with both of those components being “adjacent to” the first edge of the gate. Claim 5 also recites a “second heavily doped region” that is “adjacent to the gate” and “disposed between the drift isolation region and a device isolation region on a drain side of the device.” Each recited component is thus located relative to an edge of the gate, yielding a “source side” and “drain side” of the device with respective components on either side of the gate. Thus, the context of the claims further resolves the relational term, and each recited use of “adjacent” identifies both the structure and its reference feature (a first or second edge of the gate or a “side of the device”). GF Opening Br. at 13-14; Liu Decl. ¶¶ 49-52.

Respondents’ annotation of Figure 1 (Resp. Br. at 22-23) proves GlobalFoundries’ point. Respondents and their expert were able to locate and label every claimed structure in the figure, including the gate and its first and second edges, the channel well, the drift well, and the heavily

doped regions. The annotation includes brackets denoting the short distances between adjacent features, underscoring the nearness of those components in the Figure 1 embodiment.



Resp. Br. at 23 (annotations of components adjacent to respective gate edges)

Respondents assert that the claims “obfuscate” the meaning of “adjacent” by displacing the heavily doped region (which is adjacent to a gate edge) from the channel (which is defined in part by the same gate edge). Resp. Br. at 21. Respondents’ confusion rests on an unstated premise that “adjacent” means “contiguous or connected.” But the Federal Circuit has rejected exactly that construction. *MBO Lab’ys, Inc. v. Becton, Dickinson & Co.*, 474 F.3d 1323, 1333 (Fed. Cir. 2007) (construing “adjacent” to mean “next to” and rejecting a proposed construction of “contiguous or connected”). The plain and ordinary meaning of “adjacent” is “situated near or next.” Ex. 8 (2004 Merriam-Webster Dictionary) at 4. Because “adjacent” does not require contact, there is nothing paradoxical about structures that are near one another and also offset from one another.

Regardless, there is no contradiction in claim 1 because Respondents misread the claim. Claim 1 recites that the “first heavily doped region” is “adjacent to the first edge of the gate,” and separately that “the channel is displaced from [a/the] first heavily doped region adjacent to the first edge of the gate.” ’983 patent at cl. 1. The adjacency runs between the heavily doped region and the gate edge; the displacement runs between the *channel* and the heavily doped region. Those are two different relationships between two different pairs of structures, and Respondents’ “flat contradiction” cases are therefore inapposite. *Cf. TVnGO Ltd. (BVI) v. LG Elecs. Inc.*, 861 F. App’x 453, 459-60 (Fed. Cir. 2021); *Infinity Comp. Prods., Inc. v. Oki Data Ams., Inc.*, 987 F.3d 1053, 1060 (Fed. Cir. 2021).

Accepting that the claims allow for non-connected components to be adjacent, Respondents then demand numerical precision contrary to Federal Circuit law. Respondents argue that “a POSITA would have had no reasonable certainty for whether a distance of a picometer, a nanometer, a micrometer (microns), tens of microns, hundreds of microns, or longer is close enough to be considered ‘adjacent to’ each other.” Resp. Br. at 24-25. That is a demand for numerical precision the law does not impose. *Sonix*, 844 F.3d at 1377; *Niazi Licensing Corp. v. St. Jude Medical S.C., Inc.*, 30 F.4th 1339, 1347 (Fed. Cir. 2022). The Federal Circuit rejected an identical argument in *Ironburg*. There, the defendant contended that “elongate member” was indefinite “due to the patent’s lack of objective guidance as to how much longer than wider the member must be,” and the court held that “[s]uch numerical precision is not always required, even when using a term of degree.” 64 F.4th at 1285; *Guangdong Alison Hi-Tech Co. v. Int’l Trade Comm’n*, 936 F.3d 1353, 1359-60, 1362 (Fed. Cir. 2019) (“the degree of precision necessary [for a term of degree] is a function of the nature of the subject matter”).

Respondents’ reliance on GlobalFoundries’ invalidity contentions fails for the same reasons their reliance on the infringement contentions fails—if infringement disputes do not make a claim indefinite, neither do validity disputes. *Cf. Ironburg*, 64 F.4th at 1290. Further, Respondents misrepresent GlobalFoundries’ assertions and their own invalidity contentions. Resp. Br. at 25-26. Respondents’ contentions do not contain the colored annotations supplied in Respondents’ brief and, in fact, offer an alternative obviousness argument for this exact adjacency limitation. Ex. 10 (Ex. B4 to Resp. Prelim. Invalidity Contentions) at 32 (“To the extent there is any dispute about whether the doped drift well... is ‘adjacent to a second edge of the gate,’ it would have been obvious....”). GlobalFoundries responded that Respondents have not established an adjacency relationship from a cited passage.¹ Resp. Br. Ex. 32 at 66. GlobalFoundries’ contentions do not render the claims indefinite.

2. “the channel having a channel length that is controlled by ion implantation” (’983 patent at claims 7, 18)

Complainant’s Proposed Construction	Respondents’ Proposed Construction
Plain and ordinary meaning	“the channel having a channel length that is formed by implanting ions into the substrate to form a channel well”

Respondents’ construction reads out the requirement that the channel length “is controlled” by ion implantation and should be rejected accordingly. *Merck & Co. v. Teva Pharms. USA, Inc.*, 395 F.3d 1364, 1372 (Fed. Cir. 2005) (“A claim construction that gives meaning to all the terms of the claim is preferred over one that does not do so.”).

Respondents are unable to present any evidence that forming a channel well necessarily “control[s]” the channel length. None of Respondents’ three expert declarants offers any opinion

¹ GlobalFoundries reserves all rights to challenge the sufficiency of Respondents’ invalidity contentions.

supporting their construction of this term or the substitution of “form[ing]” a channel length in place of “controll[ing]” it. *See* Resp. Br. Ex. C. Instead, Respondents rely entirely on attorney argument that “controlling the channel well through ion implantation means implanting ions to form the channel well.” Resp. Br. at 31.

Respondents’ logic is flawed. The claims require that the channel be “define[d]” by “the first edge of the gate and the channel edge beneath the gate.” Claim 7 adds that the “length” of the channel—and thus the location of the “channel edge beneath the gate”—be “controlled by ion implantation.” But forming a channel well may not even “form” a channel length, contrary to Tower’s construction. In embodiments of the ’983 patent, the gate is formed after the channel well, such that channel, defined in part by an “edge of the gate,” may not be fully “formed” merely by ion implantation. *See* Figs. 2E-2F (gate formed after channel well 135). Substituting formation for control does not capture the plain meaning of the claim.

The specification confirms that controlling the channel length is not identical to forming a channel well. For example, the specification uses the verbs “form” and “control” to mean different things. Components (such as the channel well) are “formed,” while dimensions of those components (such as the channel length) are “controlled.” Compare ’983 patent, 8:56-57 (“the substrate is implanted with dopants **to form** a channel well 135”) with *id.*, 5:57-59 (“The effective channel length, in one embodiment, is **controlled by** ion implantation”). The specification also provides examples for controlling channel length. In embodiments, the implant is “self-aligned to the gate edge,” and the length “may be controlled by channel or body implant conditions selected to satisfy threshold voltage and breakdown voltage requirements”—for example, “shorter effective channel length may be achieved by decreasing tilt angle, implant energy or dosage and vice-versa.” ’983 patent, 5:57-67. Those are parameters that control a dimension, not the mere act of forming

a well. Additionally, it makes little sense to “form” a “channel length” since a “length” is not itself a structure of the claimed device, but rather a property of a structure. GF Opening Br. at 16-17. Respondents’ citation of “related patents” fares no better, as the cited passages reinforce that the inventors used “form” in relation to a “channel well” and not “channel length.” Resp. Br. at 34–35.

Respondents do not find any lexicography or disclaimer. Instead, Respondents allege that the prosecution history supports their construction because the Applicant was silent in the face of a claim 7 rejection despite arguing patentability of features in the independent claim. Resp. Br. at 33–34. Such silence is neither a disclaimer nor supportive of Respondents’ rewriting of the claim. *3M Innovative Props. Co. v. Avery Dennison Corp.*, 350 F.3d 1365, 1373-74 (Fed. Cir. 2002) (“An applicant’s silence in response to an examiner’s characterization of a claim does not reflect the applicant’s clear and unmistakable acquiescence to that characterization if the claim is eventually allowed on grounds unrelated to the examiner’s unrebutted characterization.”). Respondents cite *Deere* to support their view of the prosecution history, but *Deere* supports GlobalFoundries. There, the Federal Circuit rejected the patentee’s attempt to carve out certain prior art through claim construction, citing the lack of any clear disavowal in the prosecution history. *Deere & Co. v. Bush Hog, LLC*, 703 F.3d 1349, 1358-59 (Fed. Cir. 2012). That is not the case here, where Respondents seek to rewrite the claims in the face of a lack of any disavowal.

Finally, Respondents’ reliance on GlobalFoundries’ invalidity contentions fails for the same reasons discussed above. Namely, GlobalFoundries’ contentions point out the insufficiency of Respondents’ contentions, and Respondents’ brief provides new annotations not contained in their contentions.

3. “edge of the gate” (’983 patent at claims 1, 2, 6, 13-15, 20)

Complainant’s Proposed Construction	Respondents’ Proposed Construction
Plain and ordinary meaning	“edge of the gate electrode over the gate dielectric”

Respondents rewrite the claims in an attempt to read an embodiment of the specification into them. Respondents’ construction should be rejected. *K-2 Corp. v. Salomon S.A.*, 191 F.3d 1356, 1364 (Fed. Cir. 1999) (“Courts do not rewrite claims; instead, we give effect to the terms chosen by the patentee.”).

Respondents’ construction errs in two ways. First, Respondents equate the “gate” with the “gate electrode.” But the claims recite these terms separately. Namely, the gate electrode is a component of the gate. *See also* ’983 patent, 4:49-51 (“The gate includes a HV gate electrode 174.”). The “gate” is therefore broader than the “gate electrode” it includes, and substituting the component for the whole is legal error. *K-2 Corp.*, 191 F.3d at 1364. Nor do the claims require the edge of the gate be the edge of the gate electrode. Although that happens to be true in the Figure 1 embodiment, limiting the claims to that embodiment is legal error. *JVW Enters., Inc. v. Interact Accessories, Inc.*, 424 F.3d 1324, 1335 (Fed. Cir. 2005) (courts do not import limitations from embodiments “appearing only in a patent’s written description” absent a clear intent that claims and embodiments be “strictly coextensive”); *Phillips v. AWH Corp.*, 415 F.3d 1303, 1323 (Fed. Cir. 2005) (en banc) (same rule applies even “if a patent describes only a single embodiment”). A POSITA reading the specification’s description of “a gate with first and second edges 179a-b *or sidewalls*” would understand an “edge of the gate” to be a sidewall of the gate; no further construction is needed. ’983 patent, 4:46-51; GF Opening Br. at 17-18.

Second, Respondents’ construction adds “over the gate dielectric,” a limitation unsupported by any lexicography or disavowal. *Thorner v. Sony Computer Ent. Am. LLC*, 669 F.3d 1362, 1365-66 (Fed. Cir. 2012). To the extent that Respondents intend this phrase to require

that the “edge of the gate” be “over the gate dielectric,” the plain language of the claim again does not support such a reading and the Figure 1 embodiment should not be imported into the claim.

Word Count Certificate

Pursuant to Ground Rule 1.7, the undersigned certifies that this Responsive Claim Construction Brief complies with the word limitation set forth in the Ground Rule 7.2. The certified word count includes footnotes and excludes caption, tables of contents or authorities, words in figures, the word count certificate, signature block, and the certificate of service.

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CERTIFICATE OF SERVICE

It is hereby certified that copies of the foregoing **COMPLAINANT'S RESPONSIVE CLAIM CONSTRUCTION BRIEF** were served on September 3, 2026 as follows:

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EXHIBIT 9

R i v e r s i d e 

Webster's II

New College Dictionary

Webster's II

New College Dictionary



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marriage. 3. The import, significance, or thrust of something. 4. *Med.* The manner or course of healing of a surgical wound. 5. A concept regarded as the product of attention directed to an object of knowledge.

*** SYN:** INTENTION, AIM, DESIGN, END, GOAL, INTENT, MEANING, PLAN, POINT, PURPOSE, TARGET, VIEW *n.* **core meaning:** what one intends to do or achieve <was never my intention to get involved in their scheme>

in-ten-tion-al (in-tén'shə-nəl) *adj.* 1. Deliberately done <an intentional oversight> 2. Relating to logical intention or connotation.

-in-ten-tion-al-i-ty (-nāl'i-tē) *n.* **-in-ten-tion-al-ly** *adv.*

in-ter (in-tūr') *vt.* **-terred, -ter-ring, -ters.** [ME *enteren* < OFr. *enterer* < Med. Lat. *interrare* : Lat. *in-*, in + Lat. *terra*, earth.] To place in a grave or tomb.

inter- *pref.* [ME < OFr. < Lat. < *inter*, between, among.] 1. Between <international> 2. In the midst of : *within* <intertropical> 3. Mutual : mutually <interrelate> 4. Reciprocal : reciprocally <intermingle>

in-ter-a-bang (in-tēr'ə-bāŋ') *n.* *var.* of INTERROBANG.

in-ter-act (in-tār'akt') *vi.* **-act-ed, -act-ing, -acts.** To act on each other.

in-ter-ac-tion (in-tār'ak'shən) *n.* 1. *a.* The act or process of interacting. *b.* The state of undergoing interaction. 2. *Physics.* Any of four fundamental ways in which bodies can influence each other, classified as strong, weak, electromagnetic, and gravitational.

in-ter-ac-tive (in-tār'ak'tiv) *adj.* 1. Acting or capable of acting on each other. 2. *Computer Sci.* Of or relating to a two-way electronic or communications system in which response is direct and continual. 3. Of, relating to, or being a form of television entertainment in which the viewer participates directly. **-in-ter-ac-tive-ly** *adv.*

in-ter-a-li-a (in-tār'ā'lē-ə, ā'lē-ə) *adv.* [Lat.] Among other things.

in-ter-a-li-os (in-tār'ā'lē-ōs', ā'lē-ōs') *adv.* [Lat.] Among other persons.

in-ter-brain (in-tār-brān') *n.* The diencephalon.

in-ter-breed (in-tār-brēd') *v.* **-bred** (-brēd'), **-breed-ing, -breeds.** — *vi.* 1. To breed with another kind or species. 2. To breed within a narrow range or with closely related types or individuals : *inbreed*. — *vt.* To cause to interbreed.

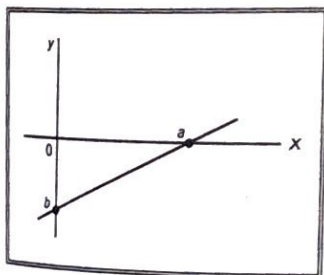
in-ter-ca-lar-y (in-tūr'kə-lər'ē, in-tār-kāl'ə-rē) *adj.* [Lat. *intercalarius* < *intercalare*, to intercalate.] 1. *a.* Inserted in the calendar to make the calendar year correspond to the solar year. — Used of a day or a month. *b.* Having an intercalary day or month inserted. — Used of a year. 2. Interpolated.

in-ter-ca-late (in-tūr'kə-lāt') *vt.* **-lat-ed, -lat-ing, -lates.** [Lat. *intercalare*, *intercalat-* : *inter-*, among + *calare*, to proclaim.] 1. To insert (a day or month) in a calendar. 2. To insert, interpolate, or interpose. **-in-ter-ca-la-tion** *n.* **-in-ter-ca-la-tive** *adj.*

in-ter-cede (in-tār-sēd') *vi.* **-ced-ed, -ced-ing, -cedes.** [Lat. *intercedere*, to intervene : *inter-*, between + *cedere*, to go.] 1. To argue on another's behalf. 2. To act as mediator in a dispute. **-in-ter-ced'er** *n.*

in-ter-cel-lu-lar (in-tār-sēl'yə-lər) *adj.* Between or among cells.

in-ter-cept (in-tār-sēpt') *vt.* **-cept-ed, -cept-ing, -cepts.** [Lat. *intercipere*, *intercept-* : *inter-*, between + *capere*, to seize.] 1. *a.* To stop or interrupt the progress or course of. *b.* To take possession of by catching (an opponent's ball), esp. in football. 2. To intersect. 3. *Obs.* To cut off from access or communication. 4. To prevent. — *n.* (in-tār-sēpt'). 1. *Math.* The distance from the origin of coordinates along a coordinate axis to the point at which a line, curve, or surface intersects the axis. 2. Interception <a radio intercept> **-in-ter-cep-tive** *adj.*



intercept

Intercept form of the equation of a line:

$$\frac{x}{a} + \frac{y}{b} = 1$$

in-ter-cep-tion (in-tār-sēp'shən) *n.* 1. The act of intercepting or state of being intercepted. 2. *a.* Something intercepted. *b.* An intercepted pass, esp. a forward pass in football.

in-ter-cep-tor also **in-ter-cept-er** (in-tār-sēp'tər) *n.* 1. One that intercepts. 2. A fast-climbing, highly maneuverable fighter plane designed to intercept enemy aircraft.

in-ter-ces-sion (in-tār-sēsh'ən) *n.* [ME < OFr. < Lat. *intercessio*, intervention < *intercessus*, p.part. of *intercedere*, to intervene. — see

INTERCEDE.] 1. Entreaty in favor of another. 2. Mediation in a dispute. **-in-ter-ces-sion-al** *adj.* **-in-ter-ces'sor** *n.* **-in-ter-ces'so-ry** *adj.*

in-ter-change (in-tār-chānj') *v.* **-changed, -chang-ing, -chang-es.** [ME *enterchaungen* < OFr. *entrechangier* : *inter-*, between < Lat. > + *changier*, to change.] — *vt.* 1. To switch each of (two things) into the place of the other. 2. To give mutually : *EXCHANGE*. — *vi.* To change places with each other. — *n.* (in-tār-chānj'). 1. The act, process, or an instance of interchanging : *EXCHANGE*. 2. A highway intersection allowing traffic to move freely from one road to another without crossing another line of traffic. **-in-ter-chang'er** *n.*

in-ter-change-a-ble (in-tār-chān'jə-bəl) *adj.* Capable of mutual exchange. **-in-ter-change'a-bil-i-ty, in-ter-change'a-ble-ness** *n.* **-in-ter-change'a-bly** *adv.*

in-ter-clav-i-cle (in-tār-klāv'ī-kəl) *n.* A bone in front of the sternum and between the clavicles in most reptiles.

in-ter-col-le-giate (in-tār-kə-lē'jīt, -jē-īt) *adj.* Involving or representing two or more colleges.

in-ter-col-um-ni-a-tion (in-tār-kə-lūm'nē-ā'shən) *n.* 1. The open spaces between the columns in a colonnade. 2. The system by which the columns of a colonnade are spaced.

in-ter-com (in-tār-kōm') *n.* [Short for INTERCOMMUNICATION.] An intercommunication system, as between two rooms.

in-ter-com-mu-ni-cate (in-tār-kə-myōō'nī-kāt') *vi.* **-cat-ed, -cat-ing, -cates.** 1. To communicate with each other. 2. To be connected or next to, as rooms. **-in-ter-com-mu-ni-ca-tion** *n.* **-in-ter-com-mu-ni-ca-tive** (-kāt'iv, -kāt'iv) *adj.*

in-ter-con-nect (in-tār-kə-nēkt') *v.* **-nect-ed, -nect-ing, -nects.** — *vi.* To be connected one to the other. — *vt.* To connect with one another. **-in-ter-con-nec-tion** *n.*

in-ter-con-ti-nen-tal (in-tār-kōn'tə-nēn'tl) *adj.* 1. Extending or occurring between or among continents <intercontinental trade> 2. Capable of traveling from one continent to another <intercontinental aircraft>

in-ter-con-ver-sion (in-tār-kən-vūr'zhən, -shən) *n.* Mutual conversion <interconversion of two chemical compounds> **-in-ter-con-vert'i-bil-i-ty** *n.* **-in-ter-con-vert'i-ble** *adj.*

in-ter-cos-tal (in-tār-kōs'təl) *adj.* [NLat. *intercostalis* : *INTER-* + Lat. *costa*, rib.] Situated or occurring between the ribs.

in-ter-course (in-tār-kōrs', -kōrs') *n.* [ME *entrecours* < OFr. *entrecours* < Lat. *intercursum*, p.part. of *intercurrere*, to mingle with : *inter-*, between + *currere*, to run.] 1. Exchange or communications between persons or groups. 2. Sexual intercourse.

in-ter-crop (in-tār-krōp') *v.* **-cropped, -crop-ping, -crops.** — *vi.* To grow a second crop between the rows of another. — *vt.* To plant a crop between the rows of (another crop). **-in-ter-crop'** *n.*

in-ter-cur-rent (in-tār-kūr'ənt) *adj.* [Lat. *intercurrents*, *intercurrent-*, p.part. of *intercurrere*, to mingle with. — see INTERCOURSE.] 1. Occurring as an interruption or delay in a process. 2. *Pathol.* Occurring during the course of an existing disease.

in-ter-den-tal (in-tār-dēn'tl) *adj.* 1. Located between the teeth. 2. Pronounced with the tip of the tongue between the teeth, as (th) in *these* or (th) in *thunder*. — *n.* An interdental consonant.

in-ter-de-pend-ent (in-tār-dī-pēn'dənt) *adj.* Mutually dependent. **-in-ter-de-pend'ence** *n.*

in-ter-dict (in-tār-dīkt') *vt.* **-dict-ed, -dict-ing, -dicts.** [Alteration of ME *entrediten*, to place under a church ban < OFr. *entredit*, an interdict < p.part. of *entredire*, to forbid < Lat. *interdicere* : *inter-*, between + *dicere*, to say.] 1. To forbid or place under an ecclesiastical or legal sanction. 2. To destroy or interrupt (an enemy line of communication) by firepower so as to halt an enemy's advance. — *n.* (in-tār-dīkt'). 1. A prohibition by court order. 2. *Rom. Cath. Ch.* An ecclesiastical censure excluding a person or area from participation in most sacraments and from Christian burial. **-in-ter-dic'tion** *n.* **-in-ter-dic'tive, in-ter-dic'to-ry** *adj.* **-in-ter-dic'tive-ly** *adv.* **-in-ter-dic'tor** *n.*

in-ter-est (in-tār-ēst, -tār-ēst, -trēst') *n.* [ME, legal claim < OFr. < Lat., it is of importance, 3rd person sing. indicative of *interesse*, to be between : *inter-*, between + *esse*, to be.] 1. *a.* A feeling of concern or curiosity about something. *b.* Something, as a quality, that causes such a feeling. 2. Regard for one's advantage or benefit : *SELF-INTEREST*. 3. *a.* A claim, right, or legal share in something. *b.* Something in which such a claim, right, or share is held. *c.* Involvement with or participation in something. 4. *a.* A charge for a loan, usu. a percentage of the amount loaned. *b.* An excess beyond what is due. 5. A group of persons sharing esp. a financial interest in an industry or enterprise. — *vt.* **-est-ed, -est-ing, -ests.** 1. To stimulate the curiosity or hold the attention of. 2. To cause to become concerned or involved with. 3. *Obs.* To concern or affect.

in-ter-est-ed (in-tār-ēst'əd, -tār-ēst'əd, -tār-ēst'əd) *adj.* 1. Having or displaying curiosity or concern. 2. Possessing a claim, right, or share. **-in-ter-est-ed-ly** *adv.* **-in-ter-est-ed-ness** *n.*

in-ter-est-ing (in-tār-ēst'ing, -tār-ēst'ing, -tār-ēst'ing) *adj.* Arousing or holding attention. **-in-ter-est-ing-ly** *adv.*

in-ter-face (in-tār-fās') *n.* 1. A surface forming a common boundary between adjacent regions. 2. *a.* A point at which independent systems or diverse groups interact. *b.* The device or system by which interaction at an interface is effected. *c.* *Computer Sci.* The point of

oo boot ou out th thin th this ũ cut ūr urge y young
yoo abuse zh vision ə about, item, edible, gallop, circus

interaction or communication between a computer and another entity, as a printer. — *v.* **-faced, -fac-ing, -fac-es.** — *vt.* 1. To join by means of an interface. 2. To serve as an interface for. — *vi.* 1. To serve as an interface or to become interfaced. 2. To coordinate or interact smoothly. — **in'ter-fa'cial** *adj.*

in'ter-fere (in'tər-fīr') *vi.* **-fered, -fer-ing, -feres.** [ME *entreferen*, to meddle < OFr. *entreferer*.] 1. To come between so as to be an impediment: **HINDER**. 2. To obstruct illegally the movement of the ball or of an opposing player, esp. to hinder illegally the catching of a pass in football. 3. To intrude in the affairs of others: **MEDDLE**. 4. To strike one hoof against the opposite hoof or leg while moving. — Used of a horse. 5. *Physics*. To produce interference with (another wave). 6. *Electron*. To prevent or inhibit clear reception of (broadcast signals). — **in'ter-fer'er** *n.* — **in'ter-fer'ing-ly** *adv.*

*** syns:** INTERFERE, MEDDLE, TAMPER *v.* *core meaning:* to involve oneself in the affairs of others <a neighbor who *interfered* in my life> Although INTERFERE and MEDDLE are interchangeable, MEDDLE is the stronger word, implying unwanted, unwarranted intrusion <*meddle* in the private lives of others> TAMPER refers to rash, harmful, or illegal intervention <a child who *tampered* with the power tools> <attempted to *tamper* with the jury>

in'ter-fer-ence (in'tər-fīr'əns) *n.* 1. *a.* An act or process of interfering. *b.* Something that interferes. 2. *a.* *Football*. The blocking of defensive tacklers to protect the ball carrier. *b.* Illegal hindrance or obstruction of the ball or of an opposing player. 3. *Physics*. The phenomenon of two or more waves of the same frequency combining to form a wave in which the disturbance at any point is the algebraic or vector sum of the disturbances due to the interfering waves at that point. 4. *Electron*. *a.* Inhibition or prevention of clear reception of broadcast signals. *b.* The distorted portion of a received signal. — **in'ter-fer-en'tial** (-fə-rēn'shəl) *adj.*

in'ter-fer-om-e'ter (in'tər-fə-rōm'ē-tər) *n.* Any of several optical, acoustical, or radio-frequency instruments that use interference phenomena between a reference wave and an experimental wave, or between two parts of an experimental wave, to determine wavelengths, wave velocities, distances, and directions. — **in'ter-fer'o-met'ric** *adj.* — **in'ter-fer'o-met'ri-cal-ly** *adv.* — **in'ter-fer-om'e'try** *n.*

in'ter-fer-on (in'tər-fīr'ōn') *n.* [INTERFER(E) + -ON³.] Any of a group of glycoproteins produced by a cell in response to viral infection and acting to prevent viral replication.

in'ter-fer-tile (in'tər-fūr'tl) *adj.* Capable of interbreeding.

in'ter-ga-lac-tic (in'tər-gə-lāk'tik) *adj.* Between galaxies.

in'ter-gen-er-a-tion-al (in'tər-jən'ə-rā'shə-nəl) *adj.* Being or occurring between generations <*intergenerational* differences>

in'ter-gla-cial (in'tər-glā'shəl) *adj.* Between glacial epochs.

in'ter-gov-ern-men-tal (in'tər-gūv'əm-mēn'tl) *adj.* Existing or occurring between two or more governments or divisions of government.

in'ter-grade (in'tər-grād') *vi.* **-grad-ed, -grad-ing, -grades.** To merge into each other through a series of stages, forms, or types. — *n.* (in'tər-grād'). A transitional grade, step, or form. — **in'ter-gra-da'tion** (-grā-dā'shən) *n.*

in'ter-group (in'tər-grōop') *adj.* Existing or occurring between two or more social groups.

in'ter-im (in'tər-īm) *n.* [< Lat., in the meantime < *inter*, at intervals, between.] An intervening period. — *adj.* Belonging to or taking place during an interim: **TEMPORARY** <an *interim* arrangement>

in'ter-i-on-ic (in'tər-ī-ōn'ik) *adj.* Located or occurring between ions.

in-te-ri-or (in-tīr'ē-ər) *adj.* [Lat., comp. of *inter*, within.] 1. Of, relating to, or located in the inside: **INNER**. 2. Of or relating to the spiritual or mental life. 3. Situated away from a shoreline or border: **INLAND**. — *n.* 1. The internal part or area. 2. One's spiritual or mental being. 3. A representation of the inside of a room or structure, as in a painting. 4. The inland part of a particular political or geographic entity. — **in-te'ri-or-i-ty** (-ōr'ē-tē, -ōr') *n.* — **in-te'ri-or-ly** *adv.*

interior angle *n.* 1. *a.* One of four angles formed between two straight lines cut by a transversal. *b.* A vertex angle measured wholly within a polygon. 2. The angle formed inside a polygon by two adjacent sides.

interior decoration *n.* The planning and execution of the layout, decoration, and furnishing of an architectural interior.

interior decorator *n.* A person who practices or specializes in interior decoration.

interior design *n.* Interior decoration. — **interior designer** *n.*

interior monologue *n.* An often lengthy monologue representing a fictional character's thoughts and emotions.

in'ter-ject (in'tər-jēkt') *vt.* **-ject-ed, -ject-ing, -jects.** [Lat. *interjicere*, *interject*-, to put between: *inter*-, between + *jacere*, to throw.] To insert between other parts or elements: **INTERPOSE**. — **in'ter-jec'tor** *n.* — **in'ter-jec'to-ry** *adj.*

in'ter-jec-tion (in'tər-jēk'shən) *n.* 1. An exclamation. 2. *a.* A part of speech made up of exclamatory words or expressions capable of standing alone. *b.* A word, phrase, or utterance used exclamatorily to express emotion, as *Heavens!* or *Oh!* — **in'ter-jec'tion-al** *adj.* — **in'ter-jec'tion-al-ly** *adv.*

in'ter-lace (in'tər-lās') *v.* **-laced, -lac-ing, -lac-es.** — *vt.* 1. To

connect by or as if by lacing together: **INTERWEAVE**. 2. To intertwine. — *vi.* To intertwine. — **in'ter-lace'ment** *n.*

in'ter-lam-i-nate (in'tər-lām'ə-nāt') *vt.* **-nat-ed, -nat-ing, -nates.** 1. To insert between layers. 2. To arrange in alternate layers. — **in'ter-lam'i-nar** (-nər) *adj.* — **in'ter-lam'i-na'tion** *n.*

in'ter-lard (in'tər-lārd') *vt.* **-lard-ed, -lard-ing, -lards.** [ME *interlarden*, to mix fat into < OFr. *entrelarder*: *entre*-, between (< Lat. *inter*-) + *lard*, *lard* < Lat. *lardum*.] To insert something foreign into

<*interlarded* the book with drawings>

in'ter-leaf (in'tər-lēf') *n.* A blank leaf inserted between two leaves of a book.

in'ter-leave (in'tər-lēv') *vt.* **-leaved, -leav-ing, -leaves.** To provide with an interleaf.

in'ter-line (in'tər-līn') *vt.* **-lined, -lin-ing, -lines.** To insert between printed lines. — **in'ter-lin'e-a'tion** (-līn'ē-ā'shən) *n.*

in'ter-line (in'tər-līn') *vt.* **-lined, -lin-ing, -lines.** To insert (a garment) with an interlining.

in'ter-lin-e-ar (in'tər-līn'ē-ər) *adj.* 1. Inserted between the lines of a text. 2. Written or printed with different languages or versions in alternating lines.

in'ter-lin-ing (in'tər-lī'nīng) *n.* An extra lining between the outer fabric and the regular lining of a garment.

in'ter-lock (in'tər-lōk') *vi.* **-locked, -lock-ing, -locks.** 1. To lock together or join closely, as by hooking or dovetailing. 2. *Comput. Sci.* To prevent initiation of new operations until current operations are completed. — *vt.* To connect in such a way so that no part can operate independently. — **in'ter-lock'** *n.*

in'ter-lo-cu-tion (in'tər-lō-kyōo'shən) *n.* [Lat. *interlocutio* < *inter*, between + *loqui*, to speak.] Conversation between two or more persons.

in'ter-loc-u-tor (in'tər-lōk'yə-tər) *n.* 1. One who participates in a conversation. 2. The performer in a minstrel show who is placed midway between the end men and engages in banter with them.

in'ter-loc-u-to-ry (in'tər-lōk'yə-tōr'ē, -tōr'ē) *adj.* Pronounced or decided during the course of a legal action and merely temporary or provisional in nature.

in'ter-lope (in'tər-lōp', in'tər-lōp') *vi.* **-loped, -lop-ing, -lopes.** [Back-formation < E. *interlope*: *inter*- + Du. *loper*, running, < MDu. < *loopen*, to run.] 1. To violate the rights, esp. the trading rights, of others. 2. To meddle in the affairs of others. — **in'ter-lop'er** *n.*

in'ter-lude (in'tər-lōod') *n.* [ME *enterlude*, a dramatic entertainment < OFr. *entrelude* < Med. Lat. *interludium*: *inter*-, between + Lat. *ludus*, play.] 1. An intervening feature, episode, or period of time. 2. *a.* A short farcical entertainment performed between the acts of a medieval morality or mystery play. *b.* A 16th-cent. genre of comedy derived from such farces. *c.* An entertainment between the acts of a play. 3. A short musical piece interposed between the parts of a longer composition.

in'ter-lu-nar (in'tər-lōō'nər) *adj.* Of or pertaining to the period between the old and new moon when the moon is not visible.

in'ter-mar-ry (in'tər-mār'ē) *vi.* **-ried, -ry-ing, -ries.** 1. To marry a member of another group. 2. To be bound together by the marriages of members. 3. To marry within one's family, clan, or tribe. — **in'ter-mar-riage** *n.*

in'ter-med-dle (in'tər-mēd'l) *vi.* **-dled, -d-ling, -dles.** [ME *entremedlen* < OFr. *entremedler*: *entre*-, between (< Lat. *inter*-) + *medler*, to mix. — see **MEDDLE**.] To intrude in the affairs of others: **MEDDLE**. — **in'ter-med'dler** *n.*

in'ter-me-di-ar-y (in'tər-mē'dē-ēr'ē) *n., pl. -ies.* 1. One who acts as a mediator. 2. One that acts as an agent between persons or things: **MEANS**. 3. An intermediate stage or state.

in'ter-me-di-ate (in'tər-mē'dē-īt) *adj.* [ME < Med. Lat. *intermediatus* < Lat. *intermedius*: *inter*-, between + *medius*, middle.] 1. Situated or occurring between two extremes or in a middle position or state. 2. Of or relating to an intermediate school. — *n.* 1. One that is intermediate. 2. An intermediary. 3. *Chem.* A substance formed as a necessary stage in the manufacture of a desired end-product. 4. An automobile smaller than a full-sized model but larger than a compact. — *vi.* (in'tər-mē'dē-ār') **-at-ed, -at-ing, -ates.** 1. To act as an intermediary: **MEDIATE**. 2. To intervene. — **in'ter-me'di-a-cy** *n.* — **in'ter-me'di-a-te-ly** *adv.* — **in'ter-me'di-a-te-ness** *n.* — **in'ter-me'di-a'tion** *n.* — **in'ter-me'di-a'tor** *n.*

intermediate school *n.* 1. A junior high school. 2. A school usually including grades 4-6.

in'ter-ment (in-tūr'mənt) *n.* The act or ritual of interring.

in'ter-mez-zo (in'tər-mēt'sō, -mēd'zō) *n., pl. -zos or -zi* (-sē, -zē) [Ital. < Lat. *intermedius*, intermediate.] 1. A brief entr'acte. 2. A short movement separating the major sections of a lengthy musical composition or work. *b.* An independent instrumental composition having the character of such a movement.

in'ter-mi-na-ble (in-tūr'mə-nə-bəl) *adj.* Boringly protracted <an *interminable* lecture> — **in'ter-mi-na-bly** *adv.*

ā pat ā pay ār care ā father ē pet ē be hw which ī tie īr pier ō pot ō toe ō paw, for oi noise ōō took

EXHIBIT 10

**Certain Semiconductor Devices, Products Containing Same, and Components Thereof,
Inv. No. 337-TA-1500
Respondents' Preliminary Invalidity Contentions – Exhibit B4**

**INVALIDITY CLAIM CHART
U.S. Patent No. 8,507,983**

U.S. Patent Application Publication No. 2008/0191291 A1 to Chen-Bau Wu (“Wu ’291”), filed April 10, 2008, published August 14, 2008, qualifies as prior art to U.S. Patent No. 8,507,983 (“the ’983 patent”) under at least 35 U.S.C. §§ 102(a) and (e).

The Asserted Claims of the ’983 patent are anticipated by Wu ’291. The Asserted Claims of the ’983 patent are also invalid under 35 U.S.C. § 103 as obvious in view of Wu ’291 alone or in combination with one or more other references identified in Respondents’ Preliminary Invalidity Contentions.¹

Claim 1

1[p] A device comprising:

To the extent that the preamble is limiting, it is disclosed or suggested by Wu ’291. For example, Wu ’291 discloses or suggests a device (e.g., a transistor):

¹ Nothing in this chart shall be treated as an admission or suggestion that Respondents agree with Complainant regarding either the scope of any of the asserted claims or the claim constructions advanced by Complainant in its infringement contentions, or that the accused products meet any limitations of the claims. The chart provides representative examples of where each element of each claim is found within the prior art, at least under Complainant’s apparent construction of the claims as applied in Complainant’s infringement contentions. The cited evidence is merely illustrative, and Respondents reserve the right to cite alternative or additional evidence. To the extent that Complainant contends that the prior art does not disclose one or more limitations of the claims, it would have been obvious to combine the teachings of the prior art with: (1) the knowledge of one of ordinary skill in the art to show all the limitations of the claims; (2) the teachings of the prior art references set forth in Respondents’ other invalidity charts with respect to the one or more limitations; and/or (3) the teachings of any of the prior art references set forth in the cover document of Respondents’ invalidity contentions, as cited herein for the one or more limitations. Because Complainant has yet to identify any limitation of the asserted claims that it contends is not fully disclosed by this prior art, either alone or in combination with other prior art cited by Respondents, Respondents expressly reserve the right to rebut any such contention, including by identifying additional obviousness combinations, if any such contention is made by Complainant. Where the chart below states that this prior art “discloses” a limitation, such disclosure may be express or inherent.

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A POSITA would have been motivated to combine Wu '291 and the other prior art given a shared focus on providing power and high-voltage semiconductor devices, particularly LDMOS devices. (E.g., Wu '291, [0002] (“The present invention relates generally to semiconductor devices and, more particularly, to high-voltage semiconductor devices.”); Levin '977, 1:9-13 (“This invention relates to integrated circuits (ICs) formed on semiconductor substrates (e.g., silicon), and more particularly to Lateral Double-Diffused Metal-Oxide-Silicon (LDMOS) transistors that are fabricated using deep sub-micron very large scale integration (VLSI) fabrication processes.”); Lee '935, [0002] (“the invention relates to power devices in which a low on-resistance can be obtained while maintaining a high breakdown voltage”), [0003]; Lin '703, [0002]; Wen '688, p. 4 (“The present invention relates to a structure of a semiconductor integrated circuit, and in particular, to a semiconductor device having a gate spacer structure with an embedded conductive structure.”); Wu '291, [0002] (“The present invention relates generally to semiconductor devices and, more particularly, to high-voltage semiconductor devices.”); Yang '928, [0002] (“The disclosed subject matter is in the field of semiconductor devices and, more particularly, power transistor semiconductor devices.”); Kuzuhara '978, [0006].)

A POSITA would have been motivated to combine Wu '291 and the other prior art given the clear similarities between the MOSFET devices they disclose. For example, Wu '291, Levin '977, Lee '935, Wen '688, Yang '928, and Kuzuhara '978 all disclose NMOS transistors in which the N-doped source is surrounded by a P-doped region that extends partially under the gate electrode. (Wu '291, Fig. 4; Levin '977, 3:34-52, Fig. 1; Lee '935, [0020], [0024], Fig. 2D; Yang '928, [0031]-[0032], Fig. 1; Kuzuhara '978, [0068], Fig. 12.) Wu '291 also discloses PMOS transistor embodiments (e.g., [0011]) similar to other prior art. (E.g., Lin '703, Fig. 6.)

A POSITA would have understood that combining Wu '291 with the other art would have required nothing more than well-known techniques applied with a reasonable expectation of success and without undue experimentation. (E.g., Levin '977, 3:41-4:27; Lee '935, [0020], [0026]-[0027]; Lin '703, [0023]-[0030]; Wen '688, pp. 7-12; Yang '928, [0034]-[0045]; Kuzuhara '978, [0071]-[0073].)

1[c] a doped drift well adjacent to a second edge of the gate.

This limitation is disclosed or suggested by Wu '291. For example, Wu '291 discloses or suggests a doped drift well (e.g., element 210 in element 104) adjacent to a second edge of the gate (e.g., the right edge of the gate):

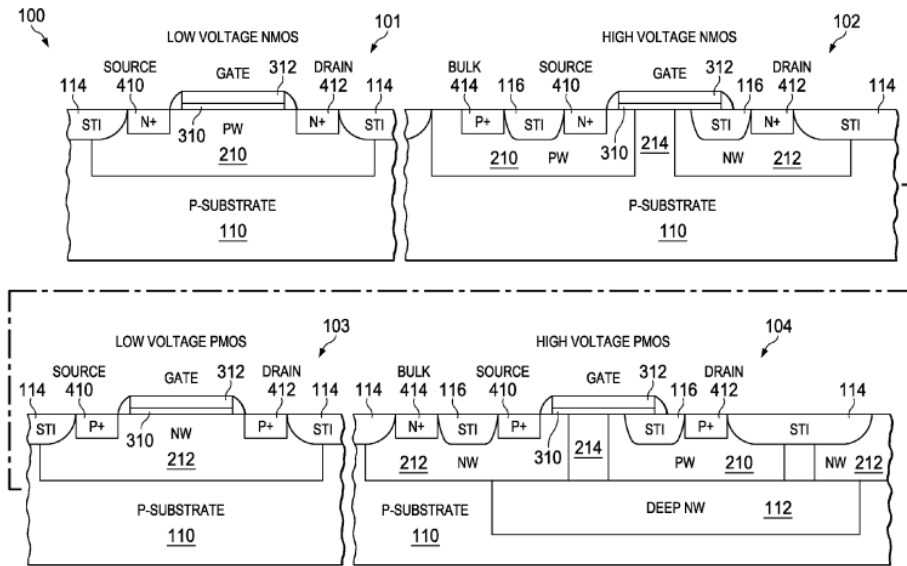


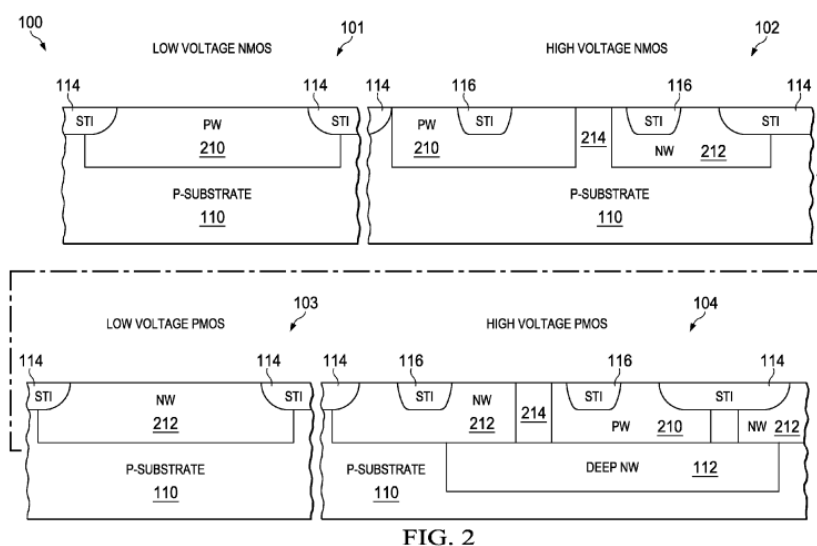
FIG. 4

(Wu '291, Fig. 4)

FIG. 4 illustrates the wafer **100** of FIG. 3 after source regions **410**, drain regions **412**, and bulk contacts **414** have been formed in accordance with an embodiment of the present invention. The source/ drain regions **410**, **412** may be formed using one or more implant processes and one or more spacers. In a preferred embodiment, the source regions **410** and drain regions **412** comprise laterally-diffused drains known in the art. However, additional doping profiles, spacers, and the like may be used. For example, the source regions **410** and drain regions **412** may comprise vertically-diffused drains, lightly-doped drains, double-diffused drains, or the like. (Examples of preferred embodiments of PMOS and NMOS devices utilizing double-diffused drains are provided in FIGS. 6, 7, 9, & 10.) The bulk contacts **414** may be formed during the same process steps as the source regions **410** and/or the drain regions **412**.

(Wu '291, [0040])

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(Wu '291, Fig. 2)

FIG. 2 illustrates the wafer **100** after P-wells **210** and N-wells **212** have been formed in accordance with an embodiment of the present invention. It should be noted that the P-wells **210** and the N-wells **212** may be doped at a lower ion concentration sufficient for low-voltage devices and are preferably simultaneously formed in the low-voltage NMOS region **101**, high-voltage NMOS region **102**, low-voltage PMOS region **103**, and high-voltage PMOS region **104**.

As described above, high-voltage devices have generally required an N-well and/or a P-well having a higher dopant concentration than the dopant concentration required for low-voltage devices. As a result, the P-wells **210** and N-wells **212** in the high-voltage PMOS and NMOS regions **102**, **104** were typically formed in separate process steps than the P-wells **210** and N-wells **212** in the low-voltage PMOS and NMOS regions **101**, **103**. Embodiments of the present invention in which the P-wells **210** and N-wells **212** are not immediately adjacent in the high-voltage PMOS and NMOS regions **102**, **104**, however, allow the high-voltage devices formed in the high-voltage PMOS and NMOS regions **102**, **104** to utilize a well region having a dopant concentration sufficient for low-voltage devices formed in the low-voltage PMOS and NMOS regions **101**, **103**. As a result, well regions for the low-voltage devices may be formed in the same process steps as well regions for the high-voltage devices, thereby reducing processing time and costs.

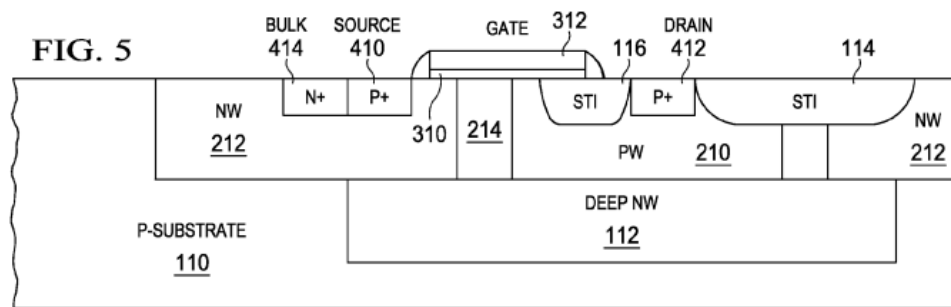
The P-well **210** may be formed by implantation with, for example, boron ions at a dose of about $1\text{E}12$ to about $1\text{E}14$ atoms/cm² and at an energy of about 50 to about 800 KeV. Other P-type dopants, such as aluminum, gallium, indium, or the like, may also be used. In an embodiment, the P-well **210** has a depth of about 0 μm to about 2 μm .

(Wu '291, [0032]-[0034])

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In accordance with an embodiment of the present invention, the P-well **210** and the N-well **212** formed within the high-voltage NMOS region **102** and the high-voltage PMOS region **104** are formed such that the P-well **210** and the N-well **212** are not immediately adjacent to each other. In other words, an interposed region **214** is positioned between the P-well **210** and the N-well **212**. The interposed region **214** is preferably a portion of the substrate **110** above the deep N-well **112**. Preferably, the interposed region **214** has a dopant concentration of about $1\text{E}14$ to about $1\text{E}17$ atoms/cm³. The distance between the P-well **210** and the N-well **212** is preferably between about $0.2\text{ }\mu\text{m}$ and $1.5\text{ }\mu\text{m}$.

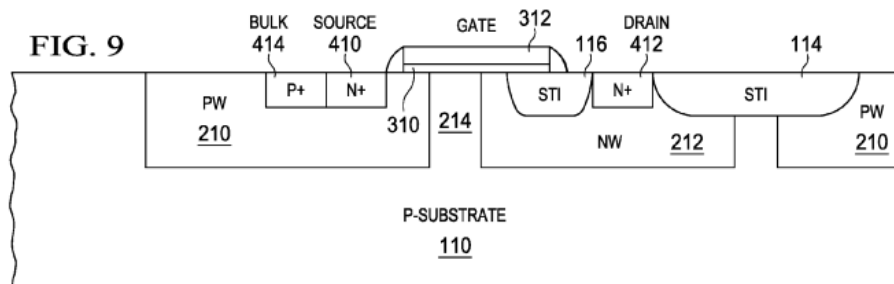
(Wu '291, [0036])



(Wu '291, Fig. 5)

FIG. 5 illustrates a lateral-diffused PMOS (LDPMOS) device having a body contact in accordance with an embodiment of the present invention. In this embodiment, the STI **116** (see FIG. 4) in the source region of the high-voltage PMOS **104** has been omitted.

(Wu '291, [0043])

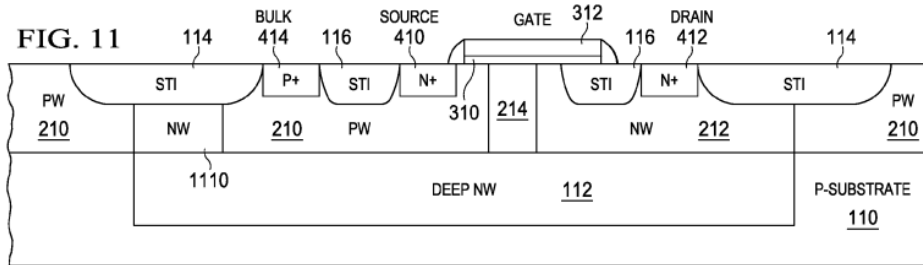


(Wu '291, Fig. 9)

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FIG. 9 illustrates a double-diffused drain NMOS (DDDNMOS) device in accordance with an embodiment of the present invention. As shown in FIG. 9, when using a double-diffused drain in an NMOS device, it may be desirable to omit the STI **116** (see FIG. 4) in the source region.

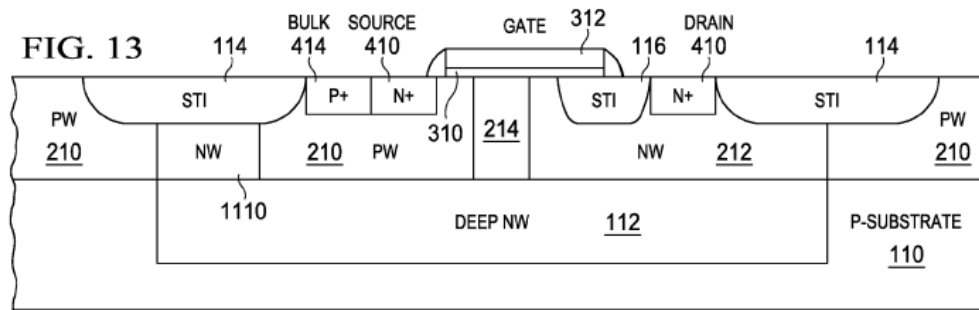
(Wu '291, [0047])



(Wu '291, Fig. 11)

FIGS. 11-14 illustrate embodiments in which a doped isolation well **1110** is used to further isolate P-well regions **210** in the source region of an NMOS device. Doped isolation well **1110** may be particularly useful in embodiments in which the potential of P-well regions **210** can be different from the potential of P-substrate region **110**. The doped isolation well **1110** may be formed simultaneously with the N-wells **212**.

(Wu '291, [0049])



(Wu '291, Fig. 13)

FIG. 13 illustrates a DDDNMOS device having a doped isolation well **1110** in accordance with an embodiment of the present invention. As shown in FIG. 13, when using a double-diffused drain in an NMOS device, it may be desirable to omit the STI **116** (see FIG. 4) in the source region.

(Wu '291, [0051])

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In accordance with an embodiment of the present invention, a high-voltage transistor is provided. The high-voltage transistor comprises a P-well and an N-well separated by a first distance, i.e., the P-well is not immediately adjacent to the N-well. A first source/drain region is located in one of the P-well and the N-well, and a second source/drain region is located in the other P-well and N-well, dependent upon whether an NMOS or a PMOS transistor is being formed.

(Wu '291, [0008])

In an embodiment, the high-voltage transistor may be fabricated by forming a first well of the first conductivity type near a surface of a semiconductor substrate, and forming a second well of a second conductivity type near the surface of the semiconductor substrate, the first well and the second well being separated by a first distance. The high-voltage transistor may then be formed such that a first source/drain region is formed in the first well and a second source/drain region is formed in the second well. One of ordinary skill in the art will appreciate that these steps may be used to fabricate either an NMOS or a PMOS high-voltage transistor.

The first and/or second well may be formed simultaneously as forming a similar well in a low-voltage region for a low-voltage transistor. In this manner, low-voltage transistors and high-voltage transistors may be fabricated in an efficient and cost-effective manner.

(Wu '291, [0011]-[0012])

Size reduction of complementary metal-oxide-semiconductor (CMOS) devices, such as transistors, has enabled the continued improvement in speed, performance, density, and cost per unit function of integrated circuits over the past few decades. As sizes are reduced, there has been a trend to integrate more functions on a single chip, some of which require higher voltage levels. The use of higher voltages with shorter gate length MOSFETs, however, may create undesirable effects, such as punch-through.

(Wu '291, [0003])

Accordingly, there is a need for high-voltage devices that may be fabricated easily and cost-effectively, particularly in conjunction with low-voltage devices.

(Wu '291, [0006])

The present invention relates generally to semiconductor devices and, more particularly, to high-voltage semiconductor devices.

(Wu '291, [0002])

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A transistor suitable for high-voltage applications is provided. The transistor is formed on a substrate having a deep well of a first conductivity type. A first well of the first conductivity type and a second well of a second conductivity type are formed such that they are not immediately adjacent each other. The well of the first conductivity type and the second conductivity type may be formed simultaneously as respective wells for low-voltage devices. In this manner, the high-voltage devices may be formed on the same wafer as low-voltage devices with fewer process steps, thereby reducing costs and process time. A doped isolation well may be formed adjacent the first well on an opposing side from the second well to provide further device isolation.

(Wu '291, Abstract)

To the extent there is any dispute about whether the doped drift well in the embodiments of Wu '291 is “adjacent to a second edge of the gate,” it would have been obvious that the position of elements at the drain edge of the gate (e.g., the doped drift well, second heavily doped region, and drift isolation region) could be adjusted such that the doped drift well is “adjacent to a second edge of the gate.” For example, it would have been obvious to combine the disclosures of the references charted herein with the knowledge of a person having ordinary skill in the art alone, and/or with the disclosures contained in any of several prior art references, including those in the Doped drift well and Drift isolation region buckets.

For example, it would have been obvious to combine Wu '291 with You '507 (U.S. Patent No. 8,119,507). You '507 discloses that the spacing of various elements on the drain side of a MOSFET can vary depending on application. For example, You '507 discloses lengths “p” and “q,” as shown below:

